

## Exercise class 7

### Linear ODE's

1. *Physical applications of separable first order ODE-s:*  
*Kreyszig 1.3 Example 6, Example 7*
2. *Linear first-order ODE-s.* Kreyszig 1.5, Problems (3), 4, (5), 9
3. *2<sup>nd</sup> order homogeneous initial value problems.* Kreyszig 2.2, Problems 17, 18
4. *Reduction of order (only for incomplete 2<sup>nd</sup> order ODE).* Kreyszig 2.1, Problems 3, (9)
5. *2<sup>nd</sup> order homogeneous with constant coefficients.* Kreyszig 2.2, Problems 22, 23, 30
6. *RLC circuits, solution of switching off protocol:*

$$\ddot{U} + \frac{R}{L}\dot{U} + \frac{1}{LC}U = 0, \quad U(0) = U_0, \quad \dot{U}(0) = 0$$

7. *Linear (in)dependence:* Verify that the functions

$$y_1 = \cos(3 \cos^{-1} x), \quad y_2 = 6x - 8x^3$$

are particular solutions of the differential equation

$$x^2 y'' - 3xy' + 3y = 0.$$

Check whether they are linearly independent by calculating the Wronskian

$$W(x) = y_1 y_2' - y_2 y_1'.$$

Evaluate  $W(x)$  for a few  $x$ . [Suggestions:  $x = 0, x = 1/2$ .]